Department of Computer Science and Enginering
University of Puerto Rico
Mayagüez Campus

CIIC 5015 – Introduction to AI
Spring 2026

Project II – Fully Connected Neural Networks
Due Date: April 01, 2025 11:59 PM

Objectives

1. Understand the design, implementation and use of deep learning models.
2. Understand the use of the fully connected neural networks for classification and regression.
3. Gain experience by implementing AI applications using modern deep learning frameworks.

Overview

You will design, implement and test a neural network model using PyTorch and Jupyter notebooks. These networks will be used to implement regression and classification tasks.

Regression

You will use the **fuel data set** to develop a model that predicts the miles per gallon (MPG) efficiency of a car based on several car attributes. The car MPG can be found in the following link:

https://archive.ics.uci.edu/ml/datasets/auto+mpg

Also, the following link shows how to manipulate the data:
https://www.tensorflow.org/tutorials/keras/regression

You can follow the same steps as in the example link above. But you will use the PyTorch and the following networks:

1. Network #1: basic regression with 1 layer and 1 neuron
2. Network #2: 4-layer network
    a. Layer 1 – 10 neurons
    b. Layer 2 – 20 neurons

    c. Layer 3 – 10 neurons
    d. Layer 4 – output neuron
3. Network #3: 5- layer network
    a. Layer 1 – 10 neurons
    b. Layer 2 – 20 neurons
    c. Layer 3 – 30 neurons
    d. Layer 4 – 20 neurons
    e. Layer 5 – output neuron

All networks will use ReLU activation in the intermediate layers, but not in the final. Train for 20 epochs with learning rate of 0.01.

In the notebook add a cell and answer the following question: Which of the three models had the least amount of error for validation?

Classification

You will use the MNIST to develop a model that predicts class for each digit. Use the following link as a template. It has a simple 3-layer implementation of a network.

https://github.com/milindmalshe/Fully-Connected-Neural-Network-PyTorch/blob/master/FCN_MNIST_Classification_PyTorch.py

You can follow the same steps as in the example above. But you will use the following networks:

1. Network #1: 4-layer network
    a. Layer 1 – 20 neurons
    b. Layer 2 – 50 neurons
    c. Layer 3 – 20 neurons
    d. Layer 4 – output neuron with softmax activation e.
2. Network #2: 6- layer network
    a. Layer 1 – 10 neurons
    b. Layer 2 – 20 neurons
    c. Layer 3 – 30 neurons
    d. Layer 4 – 20 neurons
    e. Layer 5 – 10 neurons
    f. Layer 6 – output neuron with softmax activation

3. Network #3: 6- layer network
    a. Layer 1 – 10 neurons
    b. Layer 2 – 40 neurons
    c. Layer 3 – 70 neurons
    d. Layer 4 – 40 neurons
    e. Layer 5 – 10 neurons
    f. Layer 6 – output neuron with softmax activation

All networks will use ReLU activation in the intermediate layers, but not in the final. Train for 20 epochs with learning rate of 0.01.

In the notebook add a cell and answer the following question: Which of the three models had the least amount of error for validation? How long it took to train each model?

Deliverables for Project II

You will need to provide a GitHub repo containing the following:

1) One Jupyter notebook for classification and one for regression with all the code the project, running in PyTorch 2.0

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